题目：[What Makes Training Multi-modal Classification Networks Hard?]

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机构：Facebook AI

动机
工作
overfitting-to-generalization ratio (OGR)
Gradient-Blending (G-Blend)
Theory
Practice: loss re-weighting

动机

- 相对于单模态模型，多模态模型接受更多的信息输入，应该比单模态部分强，但作者在实验中发现最好的单模态模型比多模态模型强

工作

- 作者将造成上述现象的原因分为两点：
  - 多模态模型参数量更多，更容易过拟合
  - 不同模态过拟合和泛化的速率不同，单个优化策略去训练是sub-optimal
- 作者提出**overfitting-to-generalization ratio (OGR)**去量化每个模态的过拟合现象，提出**Gradient-Blending (G-Blend)**，通过加权每个模态监督信号和joint head的监督信号，解决第二个问题，该方法在human action recognition，ego-centric action recognition， acoustic event detection上取得state-of-the-art的结果

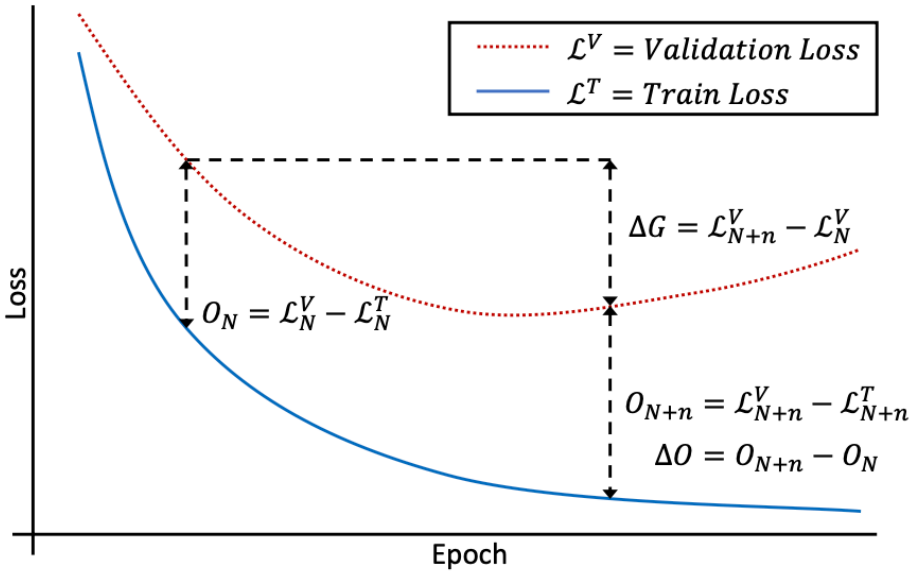
overfitting-to-generalization ratio (OGR)

$$OGR \equiv \left|\frac{\Delta O_{N,n}}{\Delta G_{N,n}}\right| = \left|\frac{O_{N+n} - O_N}{\mathcal{L}_N^* - \mathcal{L}_{N+n}^*}\right| \tag{1}$$

OGR即overfitting-to-generalization ratio，如(1)所示，它是一个量化过拟合的量。我们从下面的图进行解释：

首先是$\Delta O_{N,n}$这一项，该项等于$O_{N+n} - O_N$，表示在训练集与评估集loss的差值的变化量

然后是$\Delta G_{N,n}$这一项，该项等于$\mathcal{L}_N^* - \mathcal{L}_{N+n}^*$，$\mathcal{L}_N^*$表示了在hypothetical target distribution上真实的loss，我们认为近似等于验证集上的值。



由于两个CKPT的距离很小，其中$\hat{g}$是参数更新的梯度，我们用一范数对相关项近似：$\Delta G \approx \langle \nabla \mathcal{L}^*, \hat{g} \rangle$，$\Delta O \approx \langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, \hat{g} \rangle$，则：

$$OGR^2 = \left(\frac{\langle \nabla \mathcal{L}^{\mathcal{T}} - \nabla \mathcal{L}^*, \hat{g} \rangle}{\langle \nabla \mathcal{L}^*, \hat{g} \rangle} \right)^2 \tag{2}$$

Gradient-Blending (G-Blend)

Theory

$$w^* = \arg\min_w \mathbb{E} \left[\left(\frac{\langle \nabla \mathcal{L}^{\mathcal{T}} - \nabla \mathcal{L}^*, \sum_k w_k v_k \rangle}{\langle \nabla \mathcal{L}^*, \sum_k w_k v_k \rangle} \right)^2 \right] \tag{3}$$

作者假设下面约束成立：

$$\left\langle \nabla \mathcal{L}^*, \sum_k w_k v_k \right\rangle = 1 \tag{4}$$

那么：

$$w^* = \arg\min_w \mathbb{E} \left[\left(\left\langle \nabla \mathcal{L}^{\mathcal{T}} - \nabla \mathcal{L}^*, \sum_k w_k v_k \right\rangle \right)^2 \right] \tag{5}$$

然后计算期望：

$$
\begin{aligned}
&\mathbb{E}\left[\left(\left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, \sum_k w_k v_k \right\rangle\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_k w_k \left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, v_k \right\rangle\right)^2\right] \\
&= \mathbb{E}\left[\sum_{k,j} w_k w_j \left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, v_k \right\rangle \left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, v_j \right\rangle\right] \\
&= \sum_{k,j} w_k w_j \mathbb{E}\left[\left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, v_k \right\rangle \left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, v_j \right\rangle\right] \\
&= \sum_k w_k^2 \sigma_k^2
\end{aligned} \tag{6}
$$

其中$\sigma_k^2 = \mathbb{E}\left[\left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, v_k \right\rangle^2\right]$，在目标期望(6)上应用拉格朗日乘数法（约束(4)），则有：

$$
L = \sum_k w_k^2 \sigma_k^2 - \lambda \left(\sum_k w_k \left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle - 1\right) \tag{7}
$$

求微分：

$$
\frac{\partial L}{\partial w_k} = 2 w_k \sigma_k^2 - \lambda \left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle = 0 \tag{8}
$$

则：

$$
w_k^* = \lambda \frac{\left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle}{2\sigma_k^2} \tag{9}
$$

根据约束(4)：

$$
1 = \sum_k w_k \left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle = \sum_k \lambda \frac{\left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle^2}{2\sigma_k^2} \tag{10}
$$

则：

$$
\lambda = \frac{1}{\sum_k \frac{\left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle^2}{2\sigma_k^2}} \tag{11}
$$

设$Z = 1/\lambda$，那么:

$$
w_k^* = \frac{1}{Z} \frac{\left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle}{2\sigma_k^2} \tag{12}
$$

其中，$\sigma_k^2 \equiv \mathbb{E}\left[\left\langle \nabla\mathcal{L}^{\mathcal{T}} - \nabla\mathcal{L}^{*}, v_k \right\rangle^2\right]$ and $Z = \sum_k \frac{\left\langle \nabla\mathcal{L}^{*}, v_k \right\rangle}{2\sigma_k^2}$

Practice: loss re-weighting

$$\mathcal{L}_{blend} = \sum_{i=1}^{k+1} w_i \mathcal{L}_i \tag{13}$$

为了计算OGR，有近似$\mathcal{L}^{\mathcal{V}} \approx \mathcal{L}^*$，根据$(12)$计算$w_i = \frac{1}{Z}\frac{G^i}{{O^i}^2} = \frac{1}{Z}\frac{G_{N,n}}{{O_{N,n}}^2}$